

Curricula/Syllabi of BS Information Technology for Punjab University Affiliated Colleges

Course Title	Mobile Application Development
Course Code	EI-333
Credit Hours	3
Category	Technical Elective
Prerequisite	CC-211: Object Oriented Programming
Co-Requisite	None
Follow-up	None
Course Description	Introduction: Mobile Computing Domain, Comparison of different Mobile Platforms, Revision of Java. Android Platform: Dalvik Virtual Machine, Android Constructs (Activity, Intent, Content Provider, Service, Broadcast Receiver), Activity Lifecycle, Project Structure, Manifest File, Emulators (AVD), Debugging (DDMS), R.java, Inter Activity Communication (Intent), Event Handlers, Layout XML, @ Sign, Layouts, Adapters, Dynamic Lists, Holder Pattern, Menus and dialogs, Menu, Menu Group, Menu Item, Icon Menu, Sub Menu, Context Menu, Sub Menu, Dynamic Menu, Using XML Files for Menus, Services, Intents. Data Storage: Key Value Sets, Files, Intro to SQLite, Web Service Integration, JSON, HTTPClient, Graphics, Widgets & Notifications, Multimedia and telephony API, Android Native Development Kit (NDK).
Text Book(s)	1. Mark L. Murphy, Beginning Android 2, Apress, ISBN 978-1-4302-2629-1 2. Zigurd Mednieks, Laird Dornin, G. Blake Meike, and Masumi Nakamura Programming Android, 2nd Edition, O Reilly Press, ISBN: 978-1-449-31664-8 3. Syed Hashmi, Satya Komatineni, Dave Maclean, Pro Android 2, Apress, ISBN 978-1-4302-2659-8
Reference Material	1. http://developer.android.com/develop/index.htm 2. http://developer.android.com/guide/components/services.html 3. http://developer.android.com/guide/components/intents-filters.html 4. http://developer.android.com/guide/topics/data/data-storage.html 5. http://developer.android.com/guide/topics/graphics/index.html 6. http://developer.android.com/guide/topics/sensors/index.html